

# Transferable Mangrove Extent Mapping from Tanager-1 Hyperspectral Imagery Using Adaptive Spectral Thresholds and Pseudo-Labels

Using per-scene adaptive spectral thresholds for zero-shot transfer across four continents

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# Mangrove store outsized blue carbon, yet resist monitoring at scale

Tree obstacles stand between the sensor and a reliable, transferable map.

1

## Disproportionate blue-carbon stocks

Mangroves lock away far more carbon per hectare than most forests, making their extent a high value climate variable.

2

## A spectral resolution gap

Conventional multispectral sensors cannot reliably separate mangrove from adjacent coastal vegetation across diverse sites.

3

## Fixed thresholds do not travel

Index thresholds calibrated at one location routinely fail elsewhere (Munawaroh et al., 2025; Nie et al., 2026; Lassalle et al., 2023).



CORE IDEA

Calibrate spectral thresholds from each scene's own histogram, so no threshold, band, or label is imported from the training site. This is what enables reference-free, zero-shot transfer.

REIP and EMI use continuous red-edge and SWIR coverage that multispectral sensors lack (Lassalle 2023).

Hyperspectral bands resolve mangrove canopy structure finely enough for species-level work (Kamal & Phinn 2011; Yang 2024).

426

contiguous bands make per-scene calibration possible

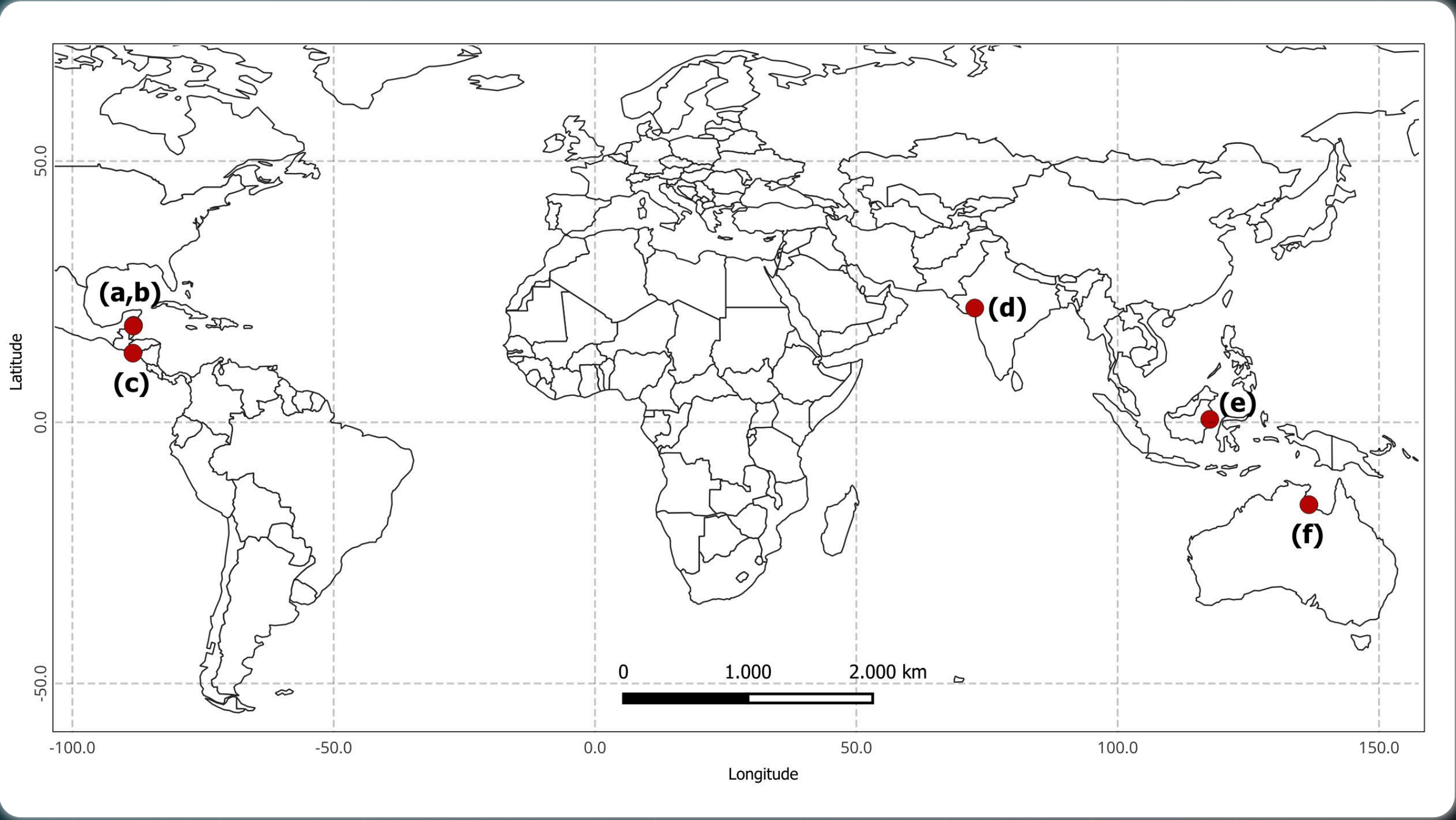
380-2500 nm  
VISIBLE-SWIR RANGE

30 m  
SPATIAL RESOLUTION

6 / 4  
SCENES / CONTINENTS

SIX SCENES, FOUR CONTINENTS, ONE TRAINING ANCHOR

Trained on one Indonesian scene,  
transferred to four continents



(a,b) Belize 1 & 2; (c) El Salvador; (d) Gujarat, India; (e) Sangatta, Indonesia; (f) Australia

| SITE           | COASTAL SETTING      | GMW V3 (HA) | ROLE            |
|----------------|----------------------|-------------|-----------------|
| Sangatta       | Tropical estuarine   | 1,216       | Training anchor |
| Australia      | Wet-dry estuarine    | 5,798       | Transfer        |
| Belize 1       | Barrier reef         | 2,730       | Transfer        |
| Belize 2       | Barrier reef         | 719         | Transfer        |
| El Salvador    | Barrier reef         | 8,893       | Transfer        |
| Gujarat, India | Semi-arid intertidal | 345         | Transfer        |

Sangatta (Indonesia) is the sole training scene; the classifier is applied to the other five without retraining.

380-2500 nm  
VISIBLE-SWIR RANGE

30 m  
SPATIAL RESOLUTION

6 / 4  
SCENES / CONTINENTS

1 → 5  
TRAINED ZERO-SHOT



# A deterministic, seven-stage pipeline

## PRE-PROCESSING

**Tanager-1**  
426 bands, 6 scenes; surface reflectance

**Feature extraction**  
8 spectral indices + REIP, per scene

**Coastal candidate mask**  
MNDWI + SAVI, Otsu <= 500m

## MODEL DEVELOPMENT

**Adaptive threshold calibration**  
Sangatta, bimodal + Otsu (MVI)

**Pseudo-label generation**  
MVI AND NDMI

**Model training**  
XGBoost, 5-fold CV, hyperparameter tuning

**GMW v3 (2020)**  
Mangrove extent as independent reference

## TRANSFER & VALIDATION

**Inference model to all 6 scenes**  
per-scene recalibration, no retraining

**Accuracy assessment**  
vs GMW v3; Kappa, F1, IoU

**Outputs**  
Mangrove extent + commission-omission + table



# Nine spectral features, two of them hyperspectral-only

## NDVI

$(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$   
Rouse (1974); Huete (1988)

## MNDWI

$(\text{Green} - \text{SWIR1}) / (\text{Green} + \text{SWIR1})$   
Xu (2006)

## NDMI

$(\text{NIR} - \text{SWIR1}) / (\text{NIR} + \text{SWIR1})$   
Gao (1996)

## CMRI

$\text{NDVI} - \text{MNDWI}$   
Gupta (2018)

## NDRE

$(\text{NIR} - \text{RedEdge}) / (\text{NIR} + \text{RedEdge})$   
Gitelson & Merzlyak (1994)

## SAVI

$(1 + L)(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red} + L)$   
Huete (1988)

## MVI

$(\text{NIR} - \text{Green}) / (\text{SWIR1} - \text{Green})$   
Baloloy (2020)

HYPER SPECTRAL

## EMI

$(\text{NIR} - \text{SWIR2}) / (\text{NIR} + \text{SWIR2})$   
Rahmila (2026)

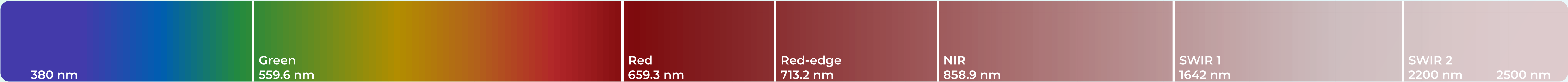
HYPER SPECTRAL

## REIP

red-edge inflection, 700–740 nm  
Guyot (1988); Cho & Skidmore (2006)

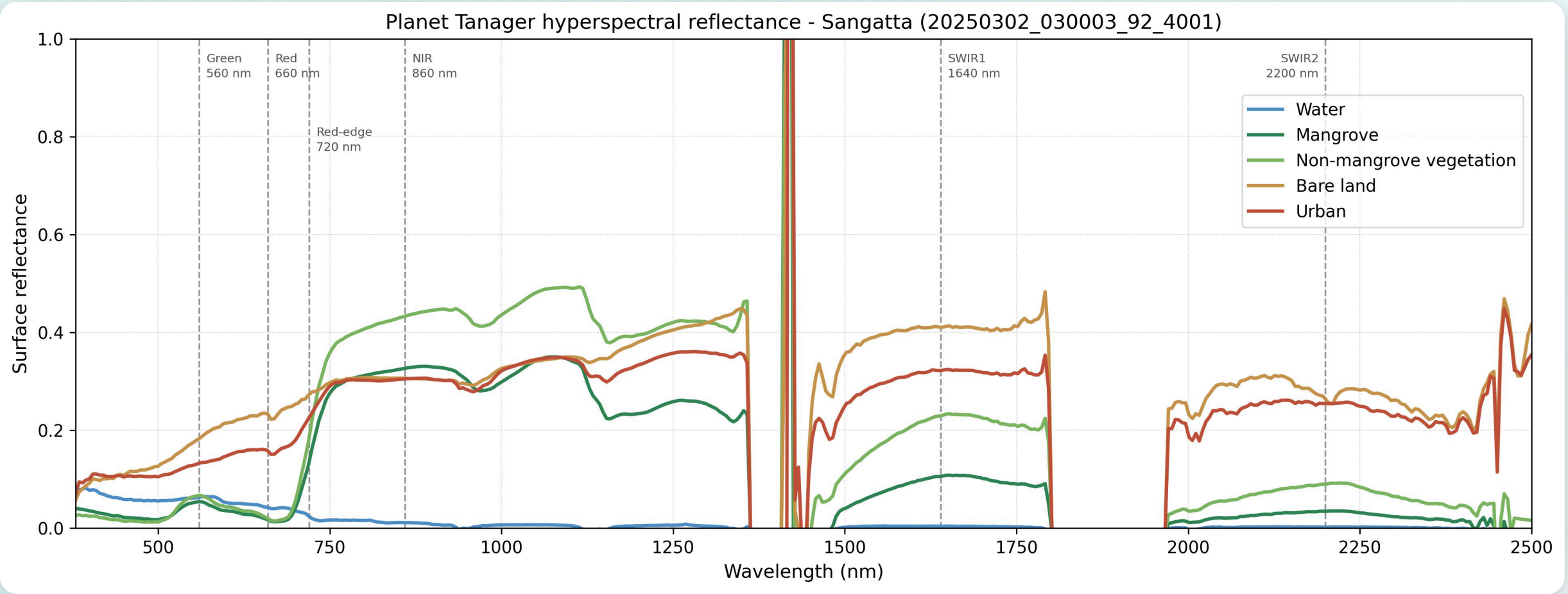


# Six bands, matched to each index’s reference wavelength



| BAND     | REFERENCE (NM) | NEAREST TANAGER BAND (NM) | FEEDS                                  | BAND-POSITION REFERENCE    |
|----------|----------------|---------------------------|----------------------------------------|----------------------------|
| Green    | 560            | 559.6                     | MNDW, MVI                              | Xu (2006)                  |
| Red      | 660            | 659.3                     | NDBI, SAVI, CMRI                       | Huete (1998)               |
| Red-edge | 720            | 719.2                     | NDRE, REIP                             | Gitelson & Merzlyak (1994) |
| NIR      | 860            | 858.9                     | NDVI, NDMI, SAVI, CMRI, NDRE, EMI, MVI | Gao (1996), Huete (1988)   |
| SWIR 1   | 1640           | 1642                      | MNDWI, NDMI, MVI                       | Xu (2006), Gao (1996)      |
| SWIR 2   | 2200           | 2200                      | EMI                                    | Rahmila (2026)             |

Tanager reconstructs 426 bands across 380 to 2500 nm; for each index the band nearest its reference wavelength is selected within a 5 nm tolerance. [src: preprocessing.py - RELEVANT\_WAVELENGTHS]



The spikes near 1400 nm are atmospheric water-vapor absorption bands, where surface signal drops to near-zero. This is standard in all VIS-SWIR hyperspectral data (EMIT, PRISMA, AVIRIS); these bands are masked before analysis. The six bands we use for indices all sit in clean atmospheric windows.



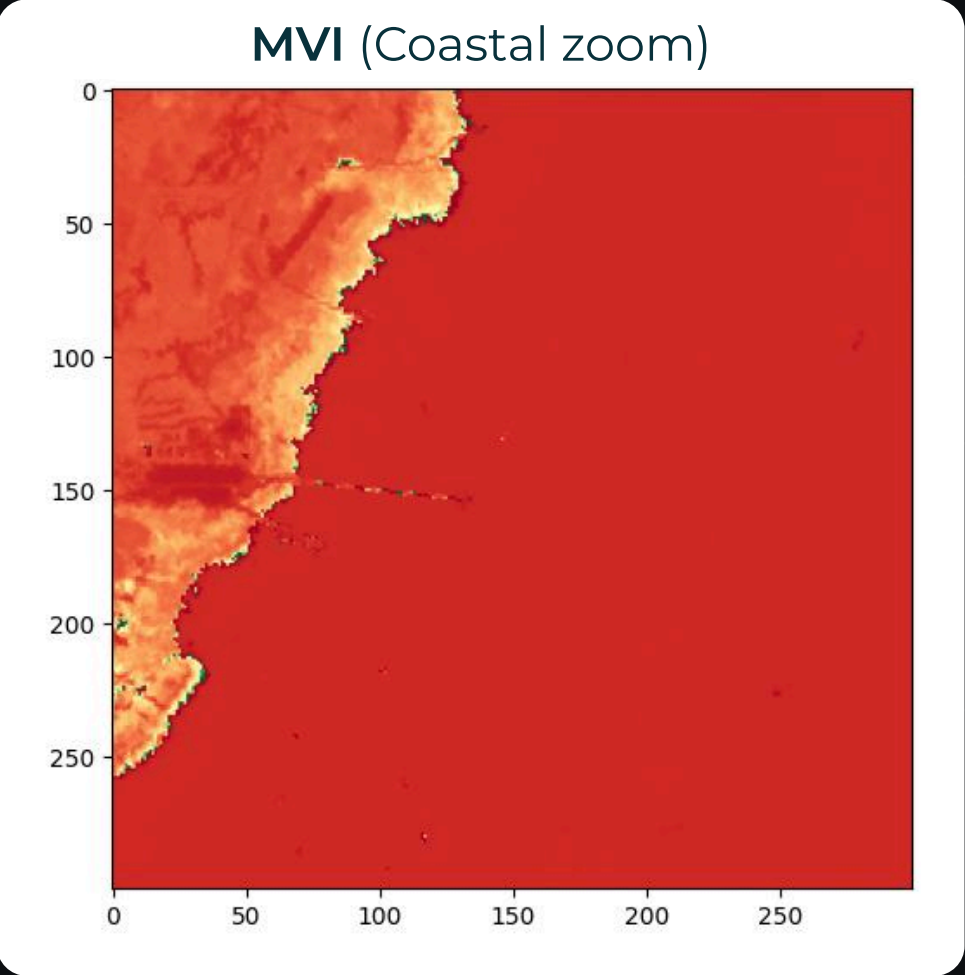
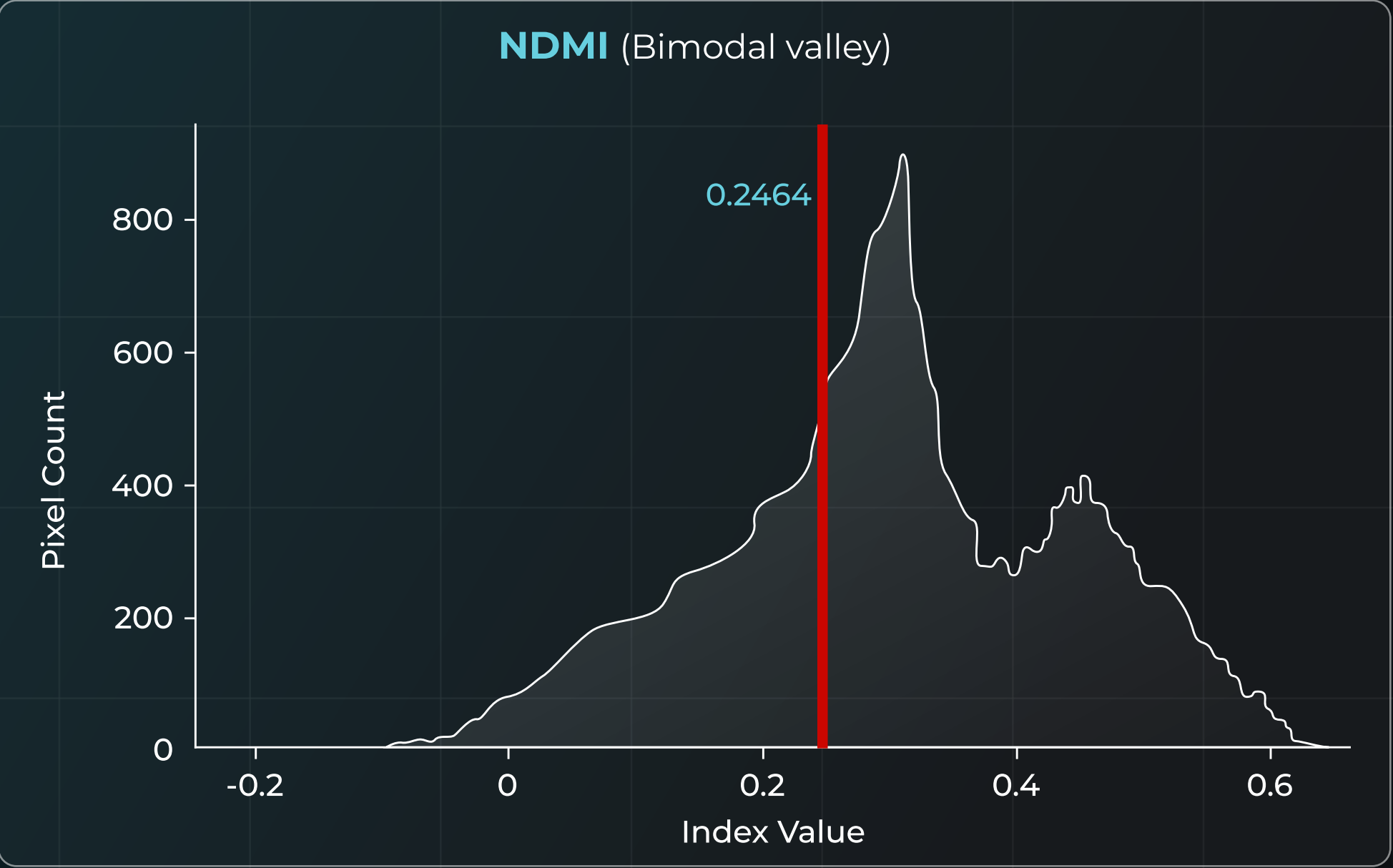
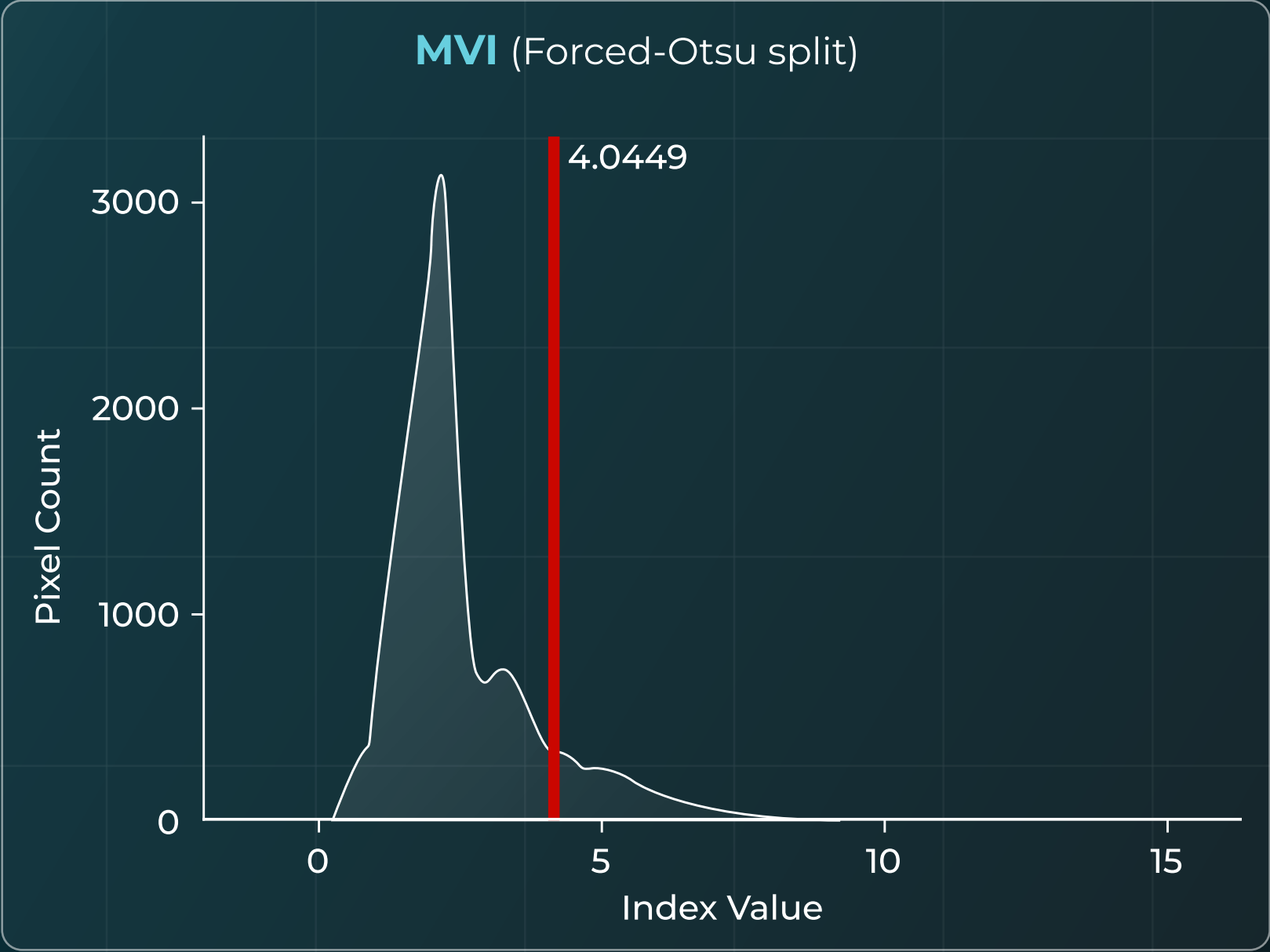
# Each scene calibrates its own thresholds

Adaptive Thresholds: Sangatta (candidate zone, n=57,014 px)

57.014 px  
CANDIDATE ZONE (N)

4.0449  
MVI THRESHOLD

0.2464  
NDMI THRESHOLD



MVI coastal zoom: the spike pixels are confined to the open-water body (right side), not the vegetated coast, confirming the artifact is a water divide-by-near-zero effect rather than a land-cover signal.

Per-scene MVI (forced-Otsu) and NDMI (bimodal valley), Sangatta candidate zone. [src: NB01 · thresholds\_sangatta.png]

- $MVI = (NIR - Green) / (SWIR1 - Green)$ : a difference ratio, not bounded to  $[-1, 1]$
- Spike near 20 = divide-by-near-zero where  $SWIR1 \approx Green$  (open water), then capped
- Far right of the threshold (4.0449) and confined to water: classification unaffected

The candidate zone is hard-capped at 500 m from water. Without the cap the Otsu buffer expanded to ~1,350 m and admitted inland (non-mangrove vegetation).

Because calibration is per-scene, nothing is imported from the training site.



# Thresholds auto-generate labels; XGBoost learns the rests

- 1

AND-logic pseudo-labels

Pixels above both MVI and NDMI thresholds become positives: 8,249 mangrove within the candidate zone.
- 2

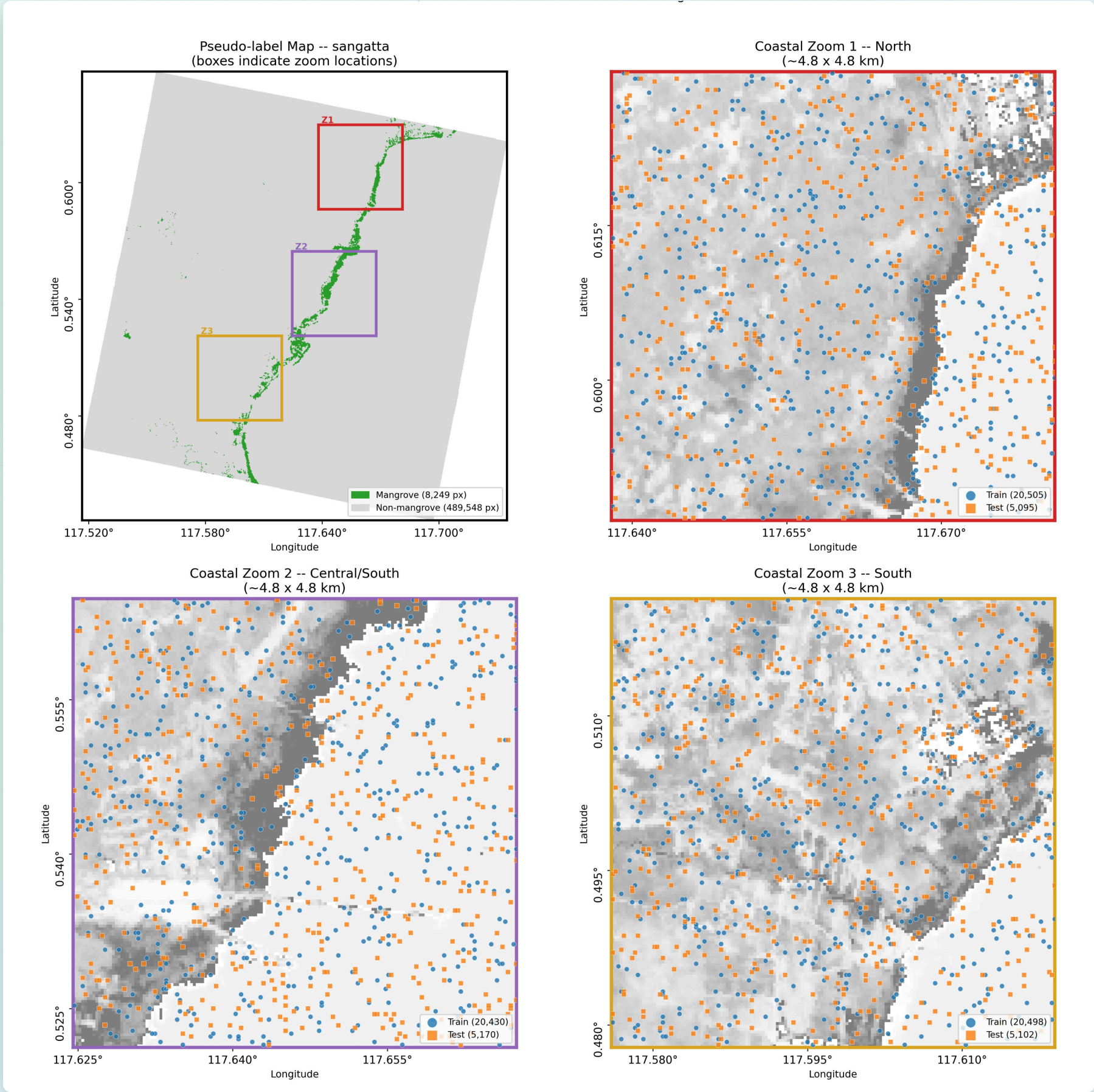
Whole-scene training matrix

497,797 pixels, split 80/20 (398,237 train / 99,560 test); XGBoost tuned by RandomizedSearchCV (50 iter, 5-fold, F1).

KEY INSIGHT

0.962<sub>CV-F1</sub> vs 0.615<sub>GMW-F1</sub>

The ceiling is pseudo-label quality, not model capacity. Effort should go to label refinement, not more tuning.



[src: NB02 - pseudo\_label\_map\_sangatta\_20250302\_030003\_92\_4001.png]



# Training-site accuracy against GMW v3 (2020)

0.887

OVERALL ACCURACY

0.548

COHEN'S KAPPA

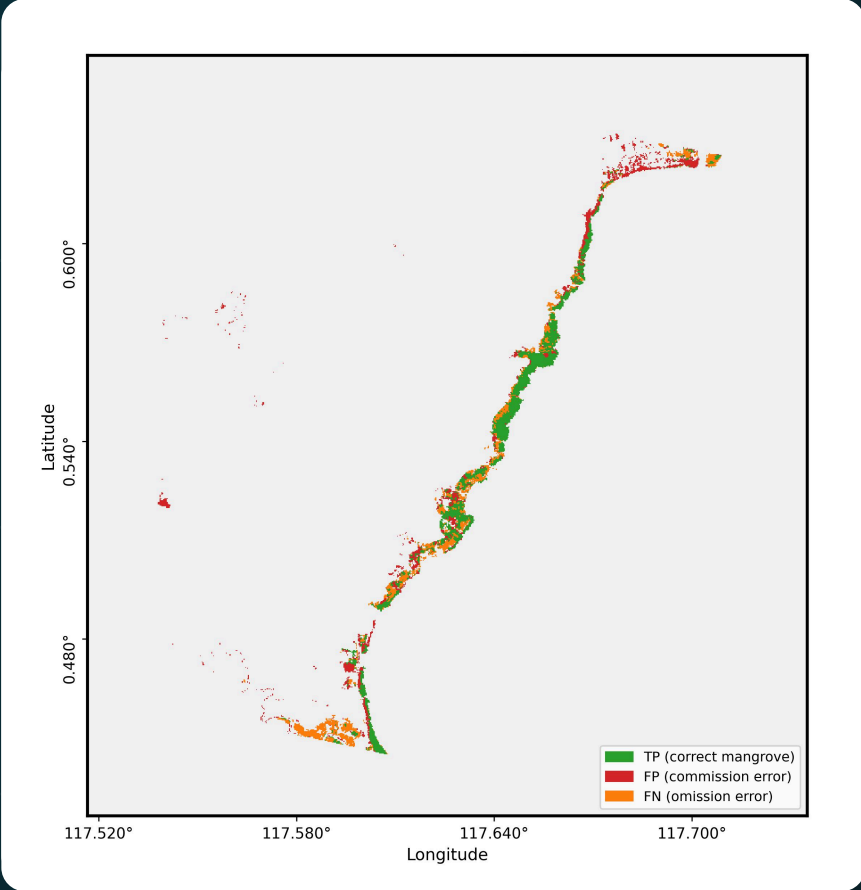
0.615

FI (MANGROVE)

0.444

IOU

GMW v3 (2020) is independent and never used in training. The 2020-vs-2025 mismatch scores five years of change as error, making Precision and Kappa conservative.



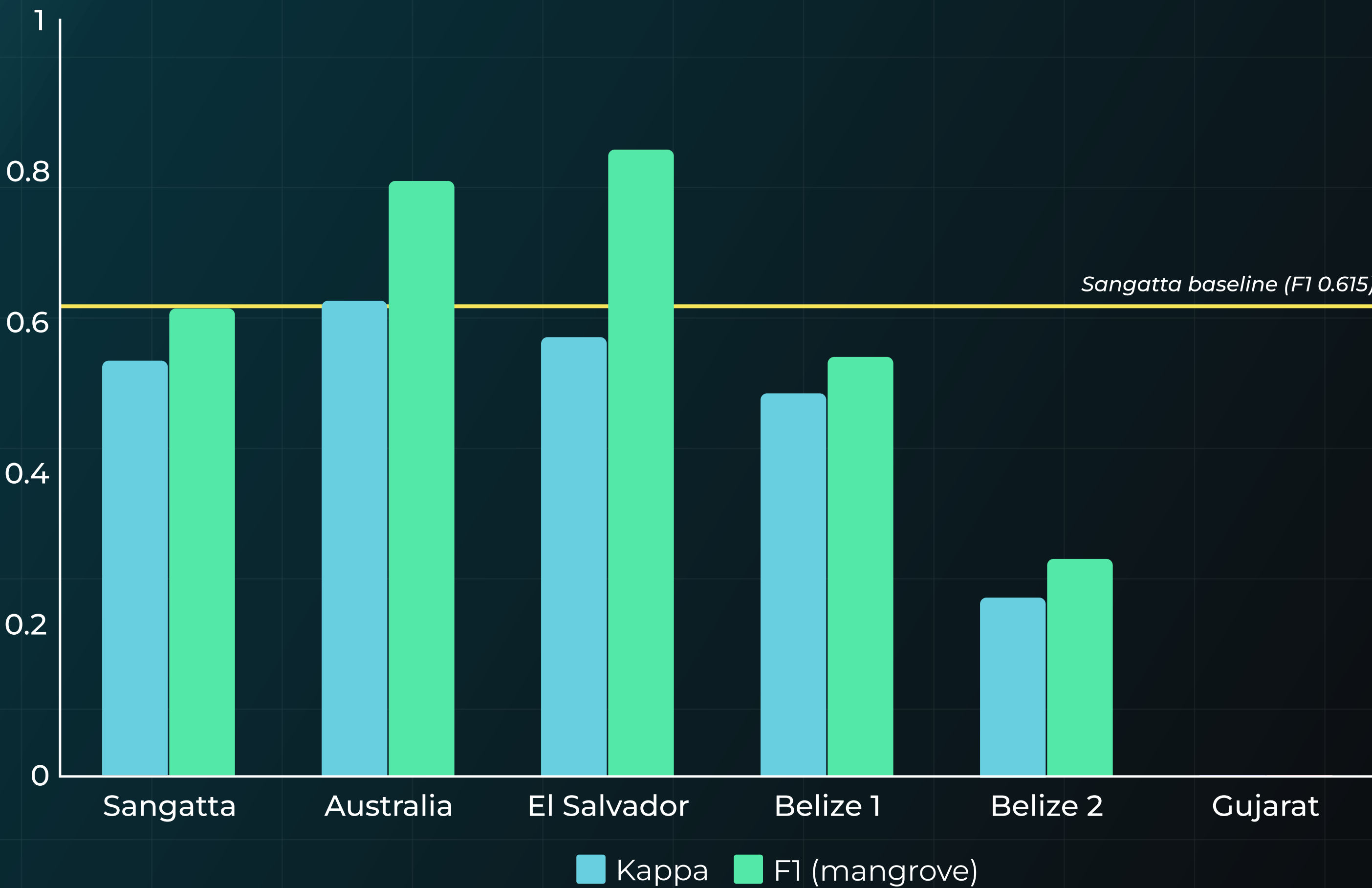
Sangatta commission-omission error map.

|                    |  | Ref: Mangrove |  | Ref: Non-mangrove |  |
|--------------------|--|---------------|--|-------------------|--|
| Pred: Mangrove     |  | 5.146<br>TP   |  | 3.091<br>FP       |  |
|                    |  |               |  |                   |  |
| Pred: Non-mangrove |  | 3.360<br>FN   |  | 45.417<br>TN      |  |
|                    |  |               |  |                   |  |

Confusion matrix, XGBoost Tuned vs GMW v3, candidate zone (57,014 px). [src: memo Table 2 - NB02 gmw\_eval CSV]



# One model, applied across four continents



1

### Generalises

El Salvador and Australia match or beat the training baseline (F1 0.83, 0.78).

2

### Over-predicts

Belize keeps high recall but low precision: true mangrove found, plus commission into coastal cover

3

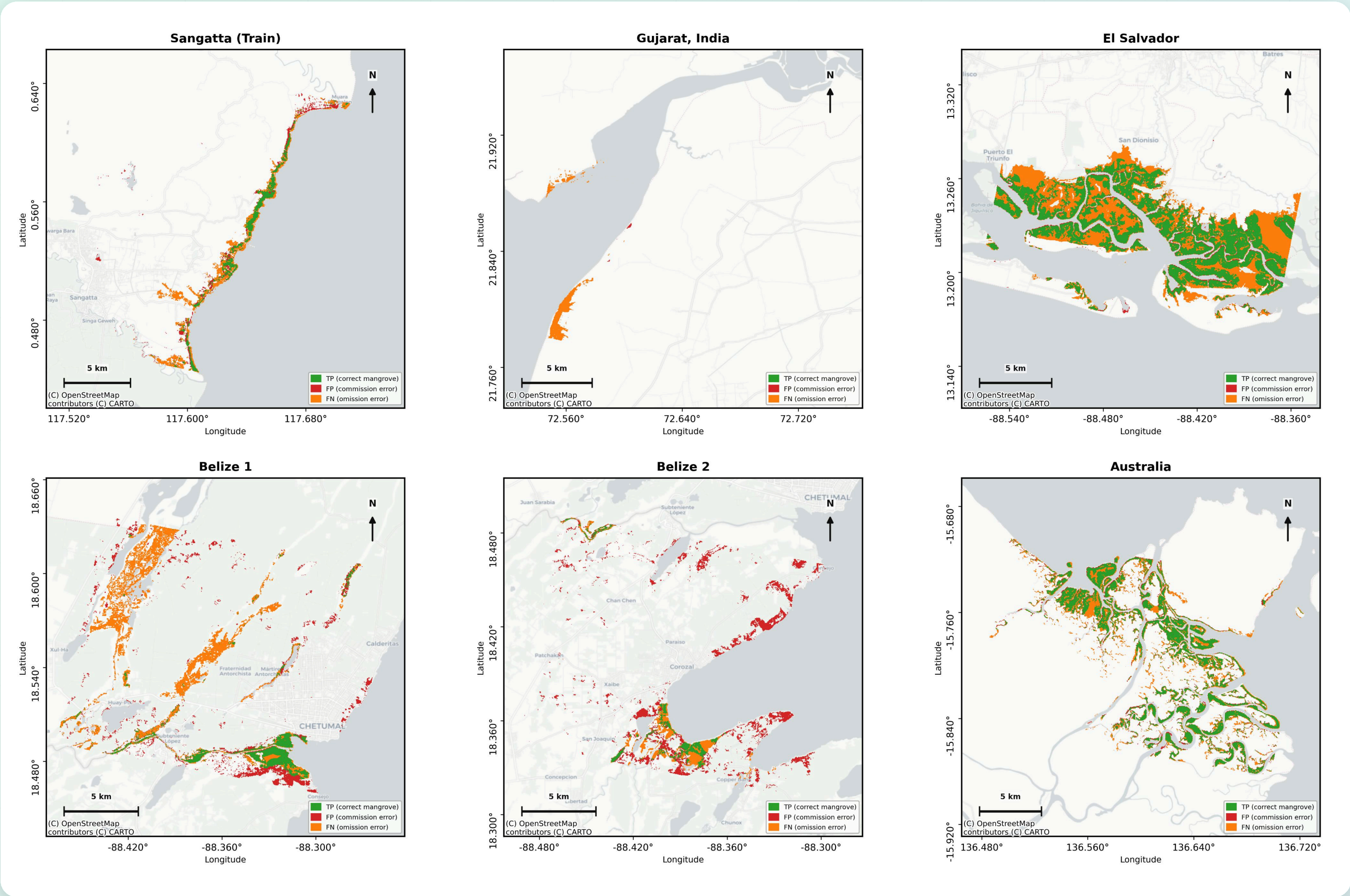
### Fails predictably

Gujarat: negligible mangrove (1.87% of the zone) leaves no mode for the threshold to find.

\*Sangatta = training baseline. [src: memo Table 3 - README results table]



# Where the errors are: commission and omission



## El Salvador and Australia stay clean

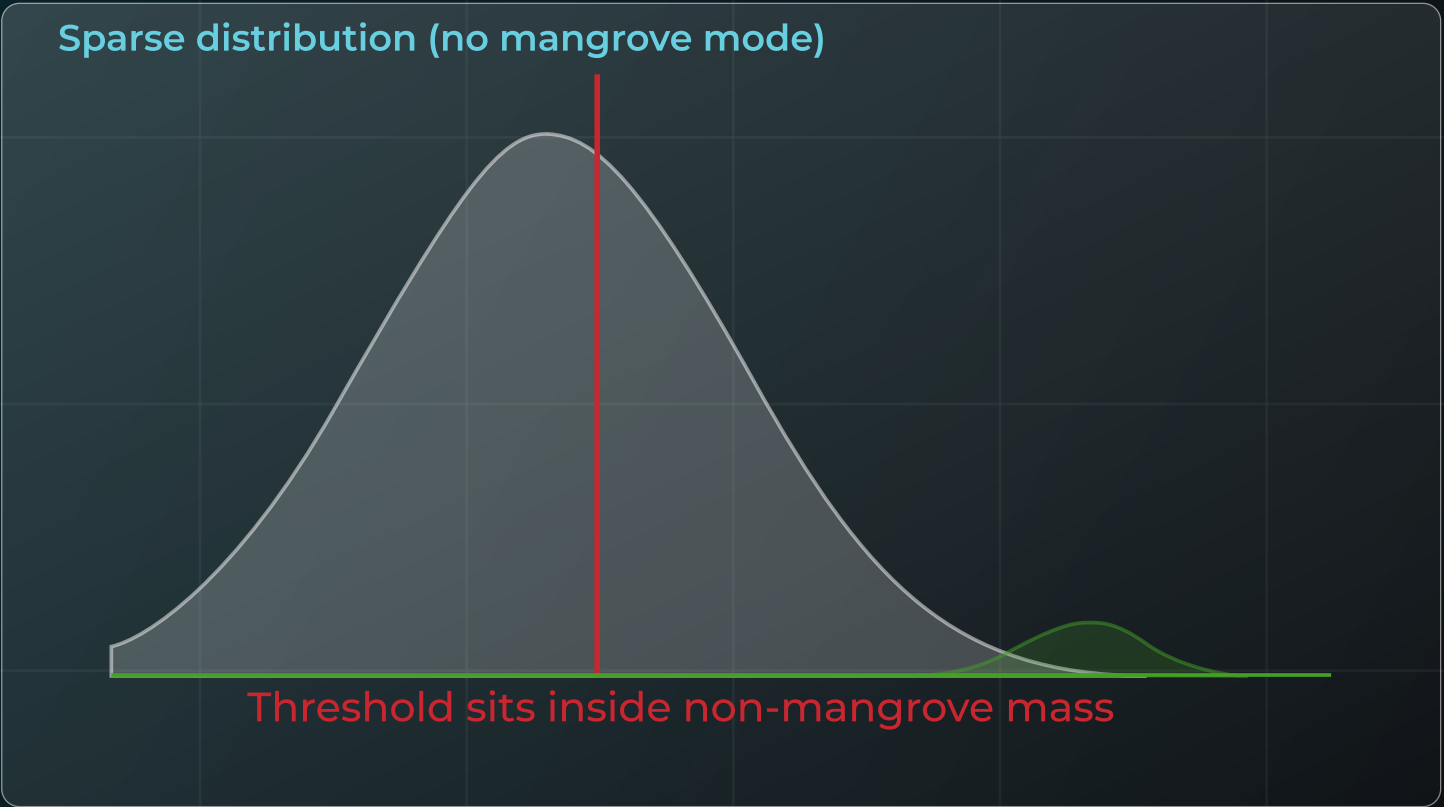
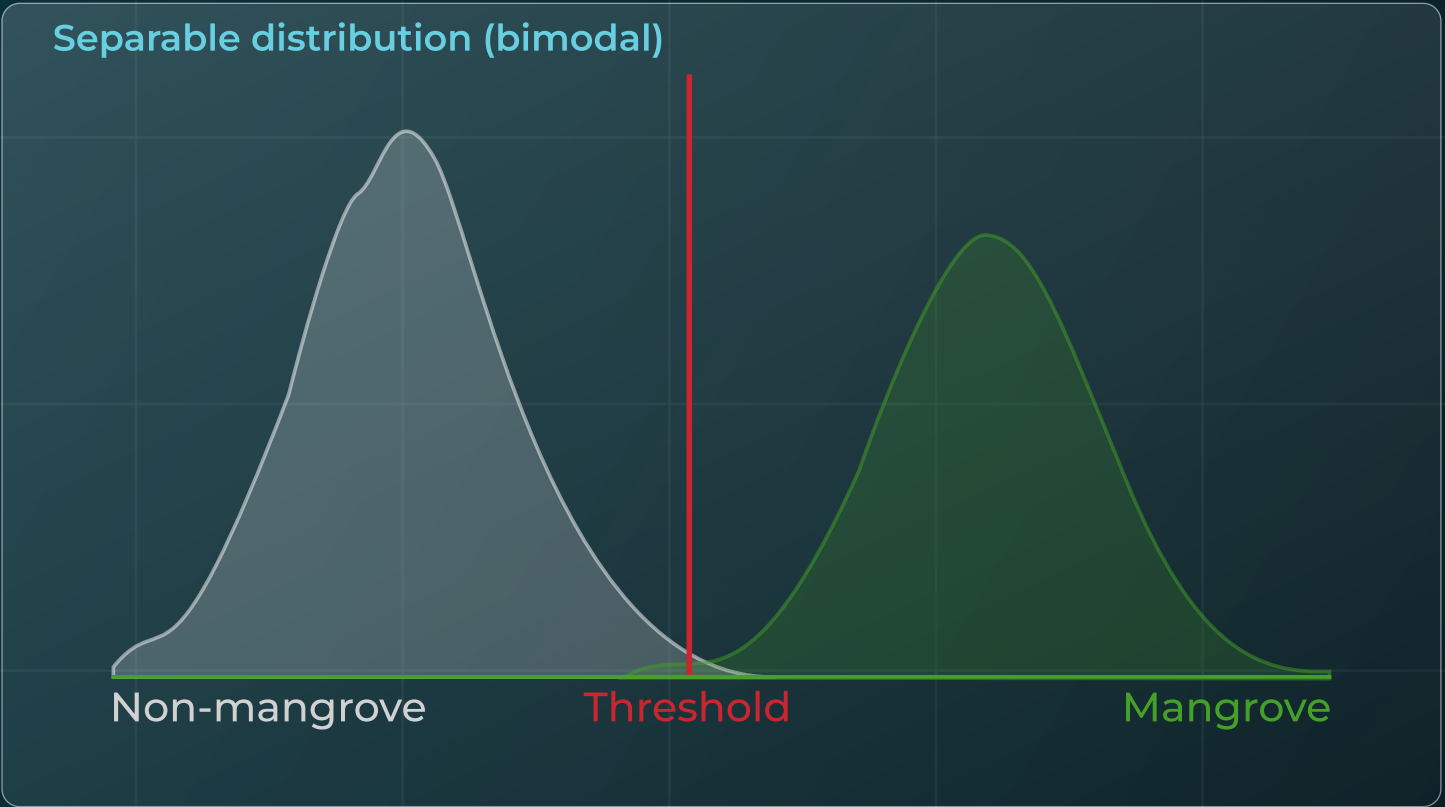
El Salvador stays in the delta (precision 0.943); Australia's inland pixels are genuine tidal-channel mangrove that GMW v3 confirms.

## Belize over-predicts inland

The candidate zone grows from every MNDWI water pixel (creeks, rivers, lagoons), and the nine features are purely spectral, with no salinity or tidal context. Freshwater and riparian vegetation therefore leak in as commission, consistent with Munawaroh (2025).



# The success/failure boundary is diagnosable



| OUTCOME | SITES                   | DIAGNOSTIC SIGNAL                             |
|---------|-------------------------|-----------------------------------------------|
| Works   | El Salvador & Australia | clear bimodal MVI/NDMI                        |
| Partial | Belize 1 & 2            | separable, but wet surroundings blur contrast |
| Fails   | Gujarat, India          | no mangrove detected (1.87% cover)            |

No field data needed: the target index distribution must be at least weakly bimodal (Nie 2026).



# Honest limits, and a field-data-free path forward

## LIMITATIONS

- **Pseudo-label ceiling:** CV-F1 0.962 far exceeds GMW-F1 0.615; tuning cannot close the gap.
- **Sparsity breaks the threshold:** complete failure at Gujarat (1.87% cover).
- **Temporal mismatch:** GMW v3 is 2020, scenes are 2025; change is scored as error.
- **Purely spectral:** no salinity or tidal context; REIP needs hyperspectral bands.

## FUTURE WORK



# A **reference-free** monitoring capability for data-scarce coasts

Hyperspectral adaptive thresholds deliver zero-shot mangrove extent without field samples, plus an interpretable account of where the method holds. It is a practical foundation for blue-carbon and biodiversity monitoring in regions that lack local training data.

## CODE (MIT)

[github.com/mwahyur46/tanager-mangrove-mapping](https://github.com/mwahyur46/tanager-mangrove-mapping)

## DATA (ZENODO)

[doi.org/10.5281/zenodo.21971757](https://doi.org/10.5281/zenodo.21971757)

## DATASET LICENSE

Tanager-1 Open STAC, CC BY 4.0



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